

Oxide-Based Metamaterials for Implant Applications: A Systematic Review and Meta-Analysis of Biochemical and Biological Performance

1. Introduction

2. Research Question

A suitable question would be:

What is the effect of oxide-based metamaterial/metamaterial-engineered implant surfaces on biochemical and biological performance compared with conventional implant materials or untreated implant surfaces?

3. Objectives

Primary objective:

- To systematically evaluate the effect of oxide-based metamaterial implant technologies on biochemical/biological performance.

Secondary objectives:

- To evaluate effects on osteogenic activity and bone formation.
- To assess cell proliferation and viability.
- To assess protein adsorption and cellular adhesion.
- To evaluate antibacterial/antimicrobial performance.
- To assess corrosion/degradation behaviour where reported.
- To identify factors influencing implant performance, such as oxide composition, surface architecture and fabrication method.

4. Methodology

4.1 Study Design

- Systematic Literature Review and Meta-Analysis
- Reporting according to **PRISMA 2020**

4.2 Databases

The literature search will be conducted in:

1. **Scopus**

2. PubMed/MEDLINE

4.3 Search Strategy

Search terms will be developed around four major concepts:

Implant + oxide material + metamaterial/metamaterial surface + biochemical/biological performance

The search strategy will be adapted to the syntax of Scopus and PubMed.

4.4 Eligibility Criteria

Inclusion:

- Studies investigating oxide-based or oxide-engineered implant materials/surfaces
- Metamaterial or metamaterial-inspired/engineered surface structures where relevant to the defined concept
- Biomedical/implant applications
- Experimental in-vitro, in-vivo or clinical studies
- Studies reporting quantitative biochemical/biological outcomes
- Original research articles
- Studies published in English

Exclusion:

- Reviews, editorials and conference abstracts where quantitative data are unavailable
- Non-biomedical applications
- Studies without relevant outcome data
- Purely theoretical/computational studies without biological or biochemical evaluation
- Duplicate publications

5. Study Selection

The PRISMA process will include:

Identification → Deduplication → Title/Abstract Screening → Full-text Assessment → Included Studies

The number of records retrieved from **Scopus and PubMed** will be reported separately and then combined after deduplication.

6. Data Extraction

A standardized extraction sheet can include:

Category	Variables
Study information	Author, year, country
Study design	In-vitro/in-vivo/clinical
Implant	Implant type and application
Material	Oxide composition
Surface	Metamaterial/nanostructured architecture
Fabrication	Coating, deposition, laser processing, etc.
Comparator	Conventional/untreated/control
Biological model	Cell line/animal model
Follow-up	Experimental duration
Biochemical outcomes	Relevant biochemical markers
Cellular outcomes	Viability, proliferation, adhesion
Osteogenic outcomes	ALP, mineralization, osteogenic markers
Antibacterial outcomes	Bacterial viability/inhibition
Other outcomes	Corrosion/degradation, if applicable

7. Risk of Bias / Quality Assessment

The assessment tool should be selected according to the final study designs.

Possible approaches:

- **SYRCLE Risk of Bias** → animal studies
- **RoB 2** → randomized clinical trials
- **ROBINS-I** → non-randomized studies
- Appropriate quality assessment for in-vitro experimental studies

If the final evidence base is predominantly laboratory studies, the risk-of-bias framework should be finalized **after screening**, rather than forcing one tool onto all study designs.

8. Outcomes for Meta-Analysis

This is particularly important because the topic is broad.

Primary outcomes

Depending on available data:

- Cell viability
- Cell proliferation
- Cell adhesion
- Osteogenic activity
- ALP activity
- Mineralization

Secondary outcomes

- Protein adsorption
- Oxidative stress/biochemical markers
- Antibacterial activity
- Bacterial adhesion
- Corrosion resistance
- Degradation behaviour
- Inflammatory response

The **final meta-analysis should only pool outcomes that are sufficiently comparable across studies.**

9. Statistical Analysis

The quantitative synthesis can include:

- Effect size calculation
- Mean Difference (MD)
- Standardized Mean Difference (SMD)
- 95% confidence intervals
- Random-effects meta-analysis

- Heterogeneity assessment
- **I² statistic**
- Cochran's Q test
- Subgroup analysis
- Sensitivity analysis
- Publication bias assessment where the number of studies permits

10. Possible Subgroup Analyses

If sufficient studies are available:

By oxide material

- Titanium oxide
- Zirconium oxide
- Aluminium oxide
- Other oxide systems

By implant application

- Dental implants
- Orthopedic implants
- Bone implants
- Other biomedical implants

By experimental model

- In-vitro
- Animal/in-vivo
- Clinical

By outcome

- Cell viability
- Cell proliferation
- Osteogenic differentiation
- Antibacterial activity

By surface modification

- Nanostructured surface
 - Coating
 - Porous architecture
 - Other engineered surface structures
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11. Results

11.1 Search Results

- Records retrieved from Scopus
- Records retrieved from PubMed
- Records after deduplication
- Records screened
- Full-text articles assessed
- Studies included

11.2 PRISMA Flow Diagram

11.3 Characteristics of Included Studies

11.4 Risk-of-Bias Results

11.5 Meta-Analysis Results

- Forest plots
- Pooled effect estimates
- Heterogeneity

11.6 Subgroup Analysis

11.7 Sensitivity Analysis

11.8 Publication Bias

12. Discussion

13. Strengths and Limitations

14. Conclusion